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Using M↓ inside TEX Documents

TUG@BachoTEX 2017

https://github.com/witiko/markdown

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Section 1

Introduction

The Case for Lightweight Markup

TEX as a Content Creation Language

1. High Markup to Text Ratio
   - *The TEXbook* (Knuth, 1986) is 22 % markup (plain TEX).
   - *Think Java* (Downey et al., 2016) is 21 % markup (LATEX).
2. Zero Sandboxing Support
   - The document you are typesetting may not compile.

     ```
     \texttt{innocent_looking_underscores.tex}
     ```

   - The document you are typesetting may halt.

     ```
     \def\whiletrue{\whiletrue} \whiletrue
     ```

   - The document you are typesetting may access the system shell.

     ```
     \immediate\write18{sudo rm -rf /}
     ```

3. Steep Learning Curve

The Case for Lightweight Markup

Comparison of LaTeX and Markdown

```latex
\section{This is a level one heading}
This is a text paragraph with \emph{emphasis}.
\begin{quotation}This paragraph will show as a quote.\end{quotation}
\begin{verbatim}
This is is a source code example.
\end{verbatim}
\begin{itemize}
  \item First item with \alert{strong emphasis}
  \item Second item with a link%
    \footnote{See \url{http://link.com} (Title)}
\end{itemize}
\begin{enumerate}
  \item First item with \verb`inline code`.
  \item Second item with an \includegraphics{image.png}
\end{enumerate}
```

The Case for Lightweight Markup

Comparison of LaTeX and Markdown

```
# This is a level one heading
This is a text paragraph with _emphasis_.
> This paragraph will show as a quote.

␣␣␣␣This is is a source code example.

* First item with **strong emphasis**
* Second item with a [link](http://link.com/ "Title")

1. First item with `inline code`.
2. Second item with an ![image](image.png "Title")
```

The Case for Lightweight Markup

Markdown as a Content Creation Language

1. Minimal Markup to Text Ratio
   - Recall: Knuth (1986) and Downey et al. (2016) are ~22 % markup.
   - *Efficient R programming* (Gillespie et al., 2016) is 5.5 % markup.
   - *R for Data Science* (Grolemund et al., 2016) is 3.8 % markup.
2. *Either* Sandboxing Support ...
   - A Markdown document converted to TEX will always compile.
   - The document may neither halt nor access the shell.
3. ... *or* Hybrid Markup Support
   - Structurally simple sections can use pure Markdown, complex sections may combine Markdown and the host markup.
4. Mild Learning Curve

Existing Solutions

The Swiss Army Knife of Pandoc

> *If you need to convert files from one markup format into another, Pandoc is your swiss-army knife.*
>
> — MacFarlane (2016b), emphasis mine

- A multi-target publishing software.
- Supports tens of markup languages (Markdown, LaTeX, HTML, XML Docbook) and output formats (ODF, OOXML, PDF).
- The use of Pandoc for the preparation of LaTeX documents has been described in TUGBoat by Dominici (2014).

Existing Solutions

Why Is Pandoc Not Ideal?

1. Difficult to Change Output Markup

```markdown
# Heading {#link}
This is [a link](#link).
```

↓↓↓

Existing Solutions

Why Is Pandoc Not Ideal?

3. Half-hybrid, Half-sandboxed
   - The input is heuristically parsed and sanitized:

     ```
     This {will} 2^n \begin{get} s~nitized and \this{will}
     not \begin{equation}2^n\end{equation} $2^n$.
     ```

     ↓↓↓

Section 2

The `markdown.tex` Package

Building a Parser

Is TEX Up to the Task?

There exist formal language parsers written solely in TEX. These parsers recognize regular (LATEX3 Project, 2016) and context-free LL(1) languages (Carlisle, 2000). Markdown is not context-free:

```
``There is a literal backtick (`) here.``
```

and a parser needs to be able to backtrack over the entire input:

```
[this is not a link](http://link.com/ "Title"
```

Implementing such a parser in TEX is possible, but generally a bad idea due to the lack of efficient data structures.

Building a Parser

Can We Use Lua Instead of TEX?

> *Lua is a powerful, efficient, lightweight, embeddable scripting language. It supports procedural programming, object-oriented programming, functional programming, data-driven programming, and data description.*
>
> — Lua Team (2016)

> *LuaTEX is an extended version of pdfTEX using Lua as an embedded scripting language.*
>
> — LuaTEX Team (2016)

Building a Parser

Can We Use Lua Instead of TEX?

- With LuaTEX, we can directly execute Lua code:

```
1 + 2 = \directlua{ tex.sprint(1 + 2) }
```

- With pdfTEX and other modern TEX engines, we can spawn a shell and execute the Lua code in a separate process:

```
1 + 2 = \newwrite\script
\immediate\openout\script=script.lua
\immediate\write\script{ print(1 + 2) }%
\immediate\closeout\script
\immediate\write18{texlua script.lua > output.tex}%
\input output.tex
```

Building a Parser

The Lunamark Library

- Lunamark (MacFarlane, 2016a) is a Markdown parser in Lua.
- The language is specified using a Parsing Expression Grammar (PEG) via the LPeg C library (and a bit of cheating).
- The dependencies of Lunamark were all either compiled into LuaTEX (LPeg, Slnunicode), or unnecessary (Cosmo, Alt-getopt).
- The library has been released under the Expat (MIT) License.

Building a Parser

A High-Level Overview

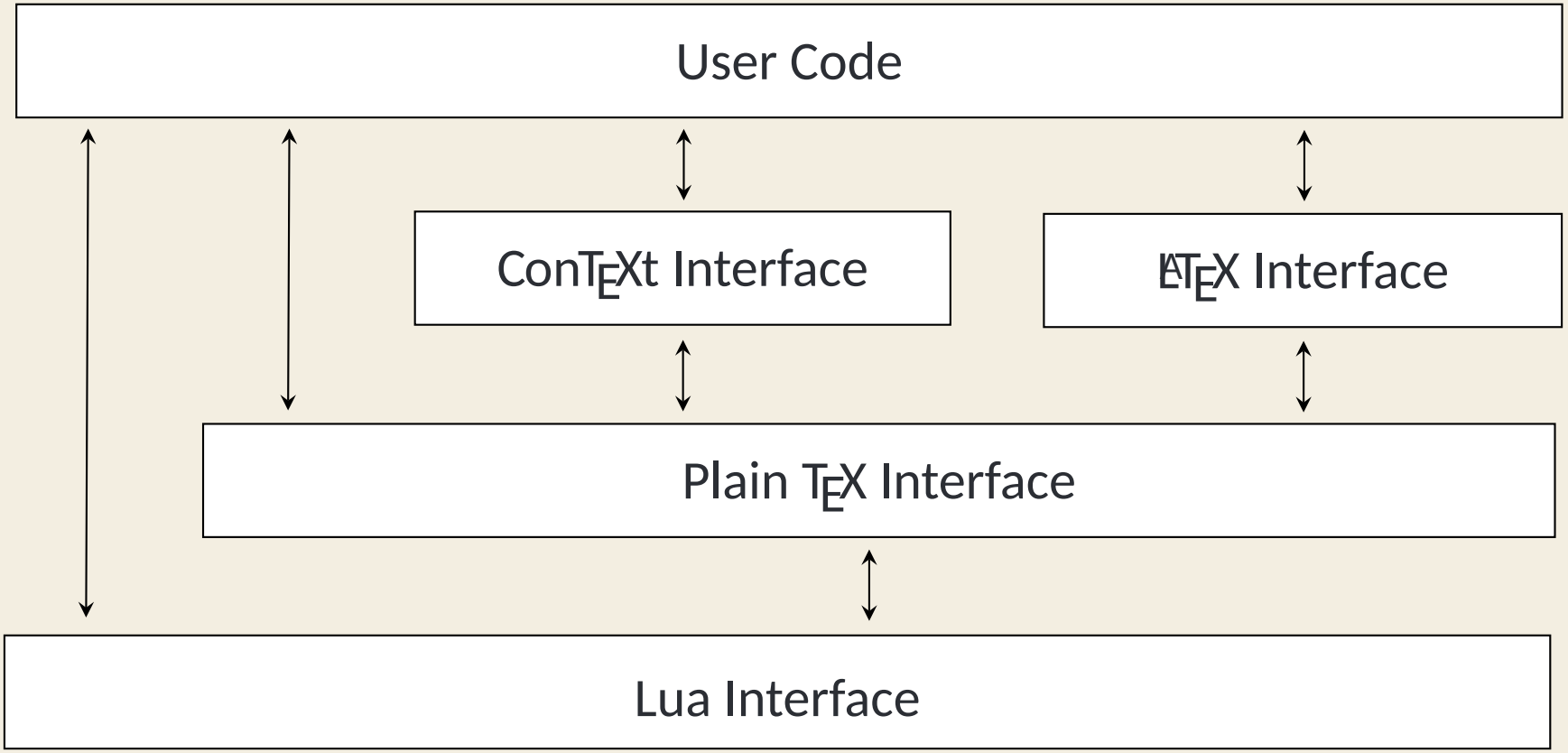
A modified version of Lunamark transforms an input Markdown document into a plain TEX document that encodes its structure:

```
# Heading
This is [a link](#link).
```

↓↓↓

Quick Overview of the Package

A Block Diagram of the Package



Quick Overview of the Package

The Lua Interface

```lua
#/usr/bin/env texlua
local kpse = require"kpse"
kpse.set_program_name"kpsewhich"
local markdown = require"markdown"
local convert = markdown.new({ something = true })
local input = "Some _Markdown_ text."
print(convert(input))
```

Quick Overview of the Package

The Plain TEX Interface

```
\input markdown %
\def\markdownOptionSomething{true}%
\def\markdownRendererSomethingElse#1#2#3{\foo{#1}}%
\markdownInput{bar.md}
\markdownBegin
Some _Markdown_ text.
\markdownEnd
\bye
```

Quick Overview of the Package

The ConTEXt Interface

```
\usemodule[t][markdown]
\def\markdownOptionSomething{true}
\def\markdownRendererSomethingElse#1#2#3{\foo{#1}}
\starttext
\markdownInput{bar.md}
\startmarkdown
Some _Markdown_ text.
\stopmarkdown
\stoptext
```

Quick Overview of the Package

The LaTeX2e Interface

```
\documentclass{article}
\usepackage[something]{markdown}
\markdownSetup{renderers = {somethingElse = {\foo{#1}}}}
\begin{document}
\markdownInput{bar.md}
\begin{markdown}
Some _Markdown_ text.
\end{markdown}
\end{document}
```

Quick Overview of the Package

The Sandbox and Hybrid Modes

```
\documentclass{article}
\usepackage{markdown}
\begin{document}
\begin{markdown}
  Foo bar \TeX{} $2^n$.
\end{markdown}
\begin{markdown*}{hybrid}
  Foo bar \TeX{} $2^n$.
\end{markdown*}
\end{document}
```

Foo bar \TeX{} 2^n. Foo bar TEX 2^n.

Quick Overview of the Package

Mapping Markdown Tokens to TEX Macros

```
\documentclass{article}
\usepackage{markdown}
\markdownSetup{renderers = {
  link = {#1\footnote{See \url{#3} (#4)}},
}}
\begin{document}
\begin{markdown}
  Foo [bar](http://link.com "Link").
\end{markdown}
\end{document}
```

Foo bar[1].

[1] See http://link.com (Link)

Quick Overview of the Package

Syntax Extensions

- Some syntax extensions were already supported by Lunamark:
  - HTML,
  - footnotes,
  - definition lists,
- New syntax extensions were added as a part of the project:
  - citations,
  - fenced code blocks,
  - IA Writer content blocks.

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{html}`

```
HTML <b>tags</b> such as &lt;b&gt; are recognized
<!-- and comments are just ignored-->.
```

HTML tags such as <b> are recognized.

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{footnotes}`

```
Here is a footnote reference,[^1] and another.[^long]

[^1]: Here is the footnote.

[^long]: Here's one with multiple blocks.

    Subsequent paragraphs are indented to show that
they belong to the footnote.
```

Here is a footnote reference,[2] and another.[3]

[2] Here is the footnote.

[3] Here's one with multiple paragraphs.

Subsequent paragraphs are indented to show that they belong to the footnote.

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{definitionLists}`

```
Term 1
:   Definition

Term 2
:   Definition with

    multiple paragraphs
```

Term 1 Definition 1

Term 2 Definition
with multiple paragraphs

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{citations}`

```
Here is a parenthetical citation [@knuth86] and
a string of several [see @knuth86, pp. 33-35;
also @gruber04, chap. 1].

Here is a text citation @knuth86 and a string of
several @knuth86 [pp. 33-35; @gruber04, chap. 1].
```

Here is a parenthetical citation (Knuth, 1986) and a string of several (see Knuth, 1986, pp. 33-35; also Gruber, 2004, chap. 1).
Here is a text citation Knuth (1986) and a string of several Knuth (1986, pp. 33-35) and Gruber (2004, chap. 1).

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{fencedCode}`

```
~~~ js
if (a > b)
  return c + 4;
else
  return d + 5;
~~~~~~
```

```
if (a > b)
  return c + 4;
else
  return d + 5;
```

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{contentBlocks}` I

```
/Flowchart.png "Engineering Flowchart"
```

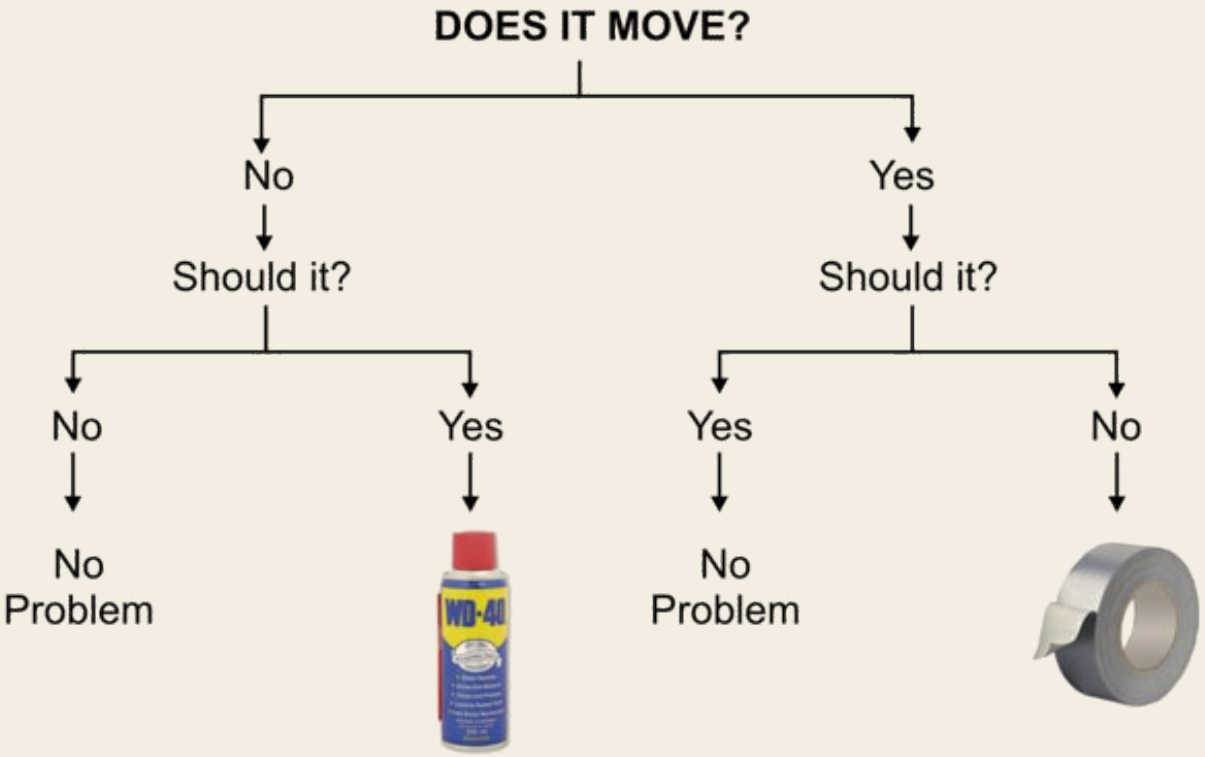


Figure: Engineering Flowchart

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{contentBlocks}` II

```
/Scientists.csv (Great Minds of the 19th Century)
```

Name	Surname	Born
Albert	Einstein	1879
Marie	Curie	1867
Thomas	Edison	1847

Table: Great Minds of the 19th Century

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{contentBlocks}` III

```
/chapters/01.txt
/chapters/02.txt
```

Chapter 1

This is the first chapter.

Chapter 2

And this is the second chapter.

Quick Overview of the Package

Syntax Extensions – `\markdownSetup{contentBlocks}` IV

```
https://tug.org/tugboat/noword.jpg
(The Communications of the \TeX{} Users Group)
```

Figure: The Communications of the TEX Users Group

(This actually does not work out-of-box at the moment.)

Section 3

Conclusion

Conclusion

The Missing Pieces of the Puzzle

The `markdown.tex` package

- enables the use of Markdown in environments where tools from outside TEX distributions are unavailable,
- gives the authors full control over how individual Markdown elements are rendered and how much access to TEX markup the Markdown documents have,
- exposes Lua, plain TEX, LATEX, and ConTEXt interfaces.
- was released under the LATEX Project Public License (LPPL) 1.3 on the Comprehensive TEX Archive Network (CTAN) and on GitHub (`https://github.com/witiko/markdown`).

Conclusion

The Missing Pieces of the Puzzle

- The syntax extensions were backported to Lunamark and merged by MacFarlane, resulting in a new minor version release of the library (0.5.0). (Novotný, 2016a)
- The package was featured on the twitter profile and the blog of Overleaf – a major online service for preparing LaTeX documents – along with original example documents. (Lim, 2017)
- TeX comments in hybrid code behave in an unexpected way.
- Logging should be improved, so that programs such as `latexmk` can automatically remove cache files that are no longer needed.

Section 4

Q&A

Section 5

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